Literature Review:

The relationship between social vulnerability and access to public transportation is a critical concern for cities striving to promote equity and inclusive urban development. Individuals considered socially vulnerable, due to factors such as low income, minority status, age, or limited access to transportation, often face heightened barriers to accessing employment, education, healthcare, and other essential services. These challenges reflect not only individual hardships but also broader structural inequalities embedded in urban transportation systems.

The Social Vulnerability Index (SVI), developed by the CDC/ATSDR, provides a valuable framework that captures factors related to socioeconomic status, household composition, minority status, housing, and transportation access. Higher SVI scores reflect greater vulnerability and often correlate with elevated risks during disruptions such as natural disasters, economic downturns, or public service interruptions – including those in transit systems (CDC/ATSDR 2021).

Complementing this framework, the Transit Desert Index (TDI) is used to identify neighborhoods where public transportation supply does not meet local demand. Areas classified as transit deserts are typically home to populations that rely heavily on public transportation but lack adequate service levels. National studies have shown that over 24 million people live in transit deserts, and that the average SVI in these areas is significantly higher – by over 20 percent – than citywide averages [1]. In some metropolitan areas, like New York–Newark–Jersey City, this disparity reaches nearly 50 percent, with poverty rates similarly elevated in underserved neighborhoods [1].

Evidence from other U.S. regions further illustrates how inequities in transit access often mirror racial and socioeconomic divides. Research in northeastern Illinois, for example, has found that transit deserts disproportionately affect Hispanic, Black, and Asian communities, particularly in areas with overlapping vulnerabilities [2]. Similarly, investigations into transit service reductions have revealed that block groups with higher poverty rates and larger Black populations are more likely to experience losses in access, especially during crises like the COVID-19 pandemic [3].

Together, these findings highlight the intersection between social vulnerability and transit accessibility. However, much of the existing literature draws on data from the early stages of the COVID-19 pandemic or from multi-city analyses that may not fully reflect current conditions in New York City. As a result, questions remain about the current state of this relationship within individual urban contexts. This study aims to build on that foundation by using recent data to examine the extent to which social vulnerability continues to align with transit access across

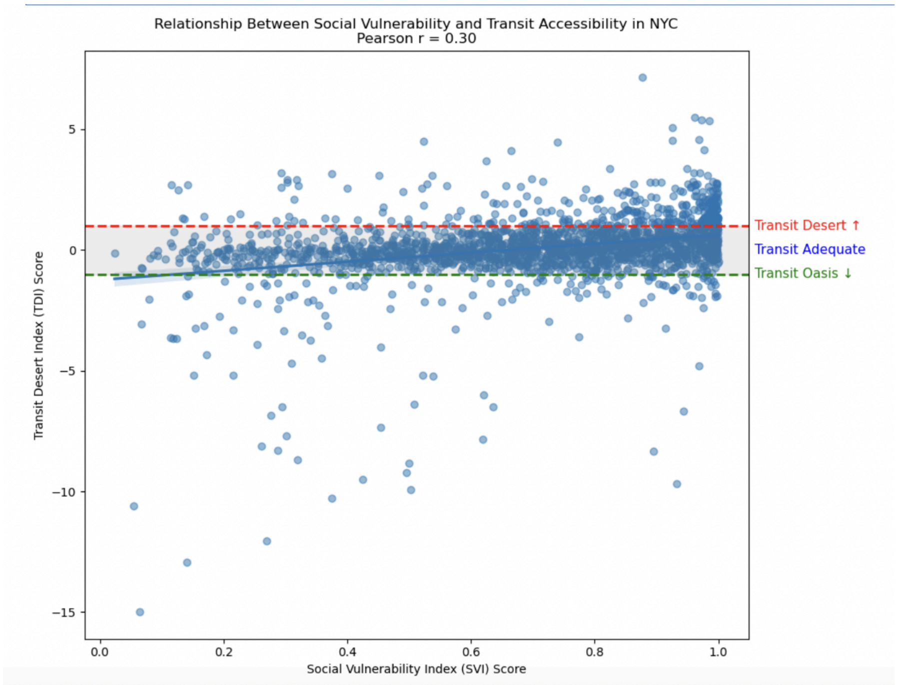
New York City census tracts. Given the city's unique infrastructure and diverse population, understanding this is significant for analyzing transit equity and informed urban planning.

Methodology

The analysis begins by loading and preprocessing two key datasets: the CDC/ATSDR Social Vulnerability Index (SVI) and the Transit Desert Index (TDI), each at the census tract level for New York City. Although other parts of this study use data measured at the census block level, the SVI is only available at the census tract, county, and ZIP Code Tabulations Area levels. Census tracts were therefore selected as the unit of analysis because they offer a practical balance – broad enough to reveal neighborhood-level patterns in transit accessibility and social vulnerability, yet granular enough to detect localized disparities. The SVI is constructed from sixteen variables sourced from the American Community survey, grouped into four themes – socioeconomic status, household characteristics, racial & ethnic minority status, and housing type & transportation – all combined into a individual and a collective percentile-based score ranging from 0 (least vulnerable) to 1 (most vulnerable).

The TDI quantifies the gap between transit demand and supply using a z-score approach. Transit demand is estimated as the population aged 12 or older, excluding institutionalized residents, divided by the area of each census tract. Transit supply is calculated by summing the z-scores of transit stops, street intersections, and points of interest within each tract. The TDI is then computed as the difference between the demand and supply z-scores, with values above 1 indicating a "transit desert" and values below -1 indicating a "transit oasis." Both datasets are filtered and joined by census tract GEOID, thus ensuring only valid tracts within the five boroughs of New York City are retained. Data cleaning steps include standardizing GEOID formats and removing tracts with missing or invalid SVI or TDI values. The merged dataset is then used to examine the relationship between overall social vulnerability and transit accessibility. The primary data product is a scatterplot of SVI versus TDI for all NYC tracts, with a fitted regression line to visualize the association.

Data Product



The scatterplot (Relationship Between Social Vulnerability and Transit Accessibility in NYC) illustrates the relationship between the Social Vulnerability Index (SVI) and the Transit Desert Index (TDI) across New York City census tracts. Each point represents a tract, with SVI on the x-axis (from least to most socially vulnerable) and TDI on the y-axis (where higher scores indicate poorer transit access). A weak to moderate positive correlation (Pearson $r = 0.30$) suggests that areas with higher social vulnerability tend to have slightly worse transit accessibility.

However, the broad dispersion of points around the trend line indicates that this relationship is completely not uniform. While many high-SVI neighborhoods do face limited transit access, others have relatively good service. Conversely, some low-SVI areas still encounter transit challenges. This variation, which contrasts with prior research showing a stronger correlation between SVI and TDI, highlights the complexity of transit equity in New York City. Additionally, it may suggest that more localized, neighborhood-level analysis may be necessary to fully understand these underlying causes and dynamics. For further details, please refer to the supplement.

Citations

(1) Jiao, Junfeng, et al. “Measuring Social Vulnerability in Transit Desert of United States Metro Areas.” *International Journal of Geospatial and Environmental Research*, vol. 8, no. 1, April. 2021, Article 3, https://ijger-ojs-txstate.tdl.org/ijger/article/view/106/7

(2) Asgharpour, Sina, et al. “Toward Equity in Accessing Public Transit: Pattern of Transit Deserts for Disadvantaged Populations.” *American Society of Civil Engineers (ASCE),* June. 2024, https://doi.org/10.1061/9780784485521.052

(3) Kar, Armita, et al. “Public transit cuts during COVID-19 compound social vulnerability in 22 US cities” *Transportation Research Part D: Transport and Environment*, vol 110, Sept. 2022, p. 103435. https://doi.org/10.1016/j.trd.2022.103435